Data Validation and Privacy Concerns with Generative AI

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Overview

Security risks and vulnerabilities

Challenges in protecting privacy

Data Poisoning

Injecting malicious or misleading data

Manipulates models' behavior and lead to biased and/or compromised outputs

Data Leakage

Sensitive or confidential information in training data or inadvertently revealed

Privacy Breaches

Fake images, text, or audio to impersonate

- Phishing attacks

Inadvertently reveal sensitive information when generating data

Privacy Breaches

Reidentification attacks

Data leakage

Security and Privacy Implications

Large datasets

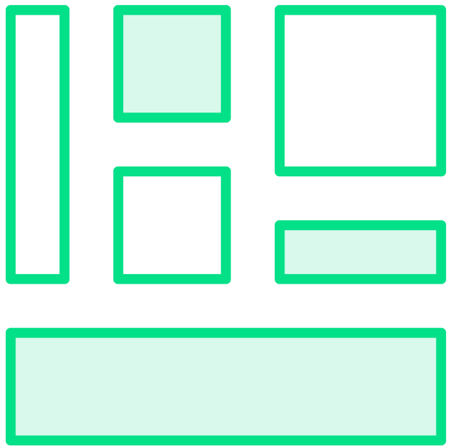
Complexity of AI algorithms

Lack of human oversight

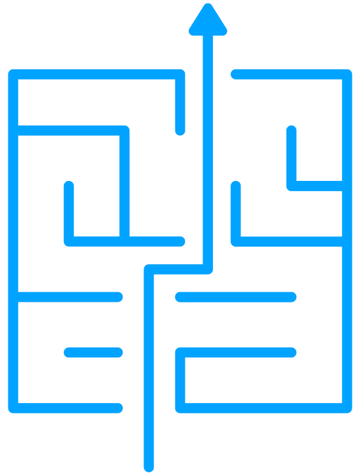
Large Datasets

Can include sensitive information, trade secrets, personal information, and more

Complexity of AI Algorithms



Complex and difficult to understand



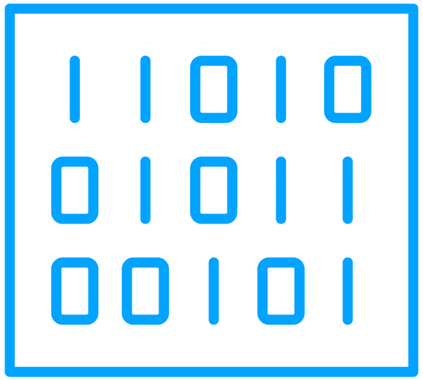
Hard to identify and mitigate vulnerabilities

Lack of Human Oversight

Lack of oversight when assessing vulnerabilities or monitoring output

Data curation leads to biased or inappropriate data in the training set

Implications on Organizations



Sensitive company data

Intellectual property (IP)

Implications on Organizations

Loss of competitive advantage

Financial loss

Reputational damage

Non-compliance

Impact on Cybersecurity

Deepfakes

Synthetic Data

Deepfakes

Manipulated audio, video, or images

Social engineering and phishing attacks

Spreading false information

Misleading evidence

Synthetic Data

Data that is artificially generated

Bypass controls

Large Datasets

Ensure data is accurate, complete, and consistent

Risks of Lack of Data Validation

False assumptions

Discrimination

Bias

Hallucinations and fabrications

Data poisoning

Challenges

Amount of data

Sophistication of data analysis techniques

Lack of awareness